

	Any Medium	Lossless Medium ($\sigma = 0$)	Low-Loss Medium ($\epsilon''/\epsilon' \ll 1$)	Good Conductor ($\epsilon''/\epsilon' \gg 1$)	Units
$\alpha =$	$\omega \left[\frac{\mu_0 \epsilon' \epsilon_0}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} - 1 \right] \right]^{1/2}$	0	$\frac{\pi \epsilon''}{\lambda_0 \sqrt{\epsilon'}}$	$\frac{\pi \sqrt{2 \epsilon''}}{\lambda_0}$	(Np/m)
$\beta =$	$\omega \left[\frac{\mu_0 \epsilon' \epsilon_0}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'} \right)^2} + 1 \right] \right]^{1/2}$	$\frac{2\pi \sqrt{\epsilon'}}{\lambda_0}$	$\frac{2\pi \sqrt{\epsilon'}}{\lambda_0}$	$\frac{\pi \sqrt{2 \epsilon''}}{\lambda_0}$	(rad/m)
$\eta_c =$	$\sqrt{\frac{\mu_0}{\epsilon' \epsilon_0}} \left(1 - j \frac{\epsilon''}{\epsilon'} \right)^{-1/2}$	$\frac{\eta_0}{\sqrt{\epsilon'}}$	$\frac{\eta_0}{\sqrt{\epsilon'}}$	$\frac{(1 + j) \eta_0}{\sqrt{2 \epsilon''}}$	(Ω)
$u_p =$	ω / β	$c / \sqrt{\epsilon'}$	$c / \sqrt{\epsilon'}$	$c \sqrt{2 / \epsilon''}$	(m/s)
$\lambda =$	$2\pi / \beta = u_p / f$	u_p / f	u_p / f	u_p / f	(m)

Notes: In practice, a material is considered a low-loss medium if $\epsilon''/\epsilon' < 0.01$ and a good conducting medium if $\epsilon''/\epsilon' > 100$; $c = 3 \times 10^8$ m/s; $\eta_0 = 377 \Omega$.